# Lenin and Stalin on Russian TV

Forecasting historical references in a television archive

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## Forecasting problem

How many references to Lenin and Stalin will appear on four federal TV channels over the next 31 days?

## Why should we care?

Study political narratives through historical references.
Track changes over time and compare channels.
Build a profile of editorial framing and examine possible propagandistic uses of historical figures.

## Forecast specification

### TARGET VARIABLES

Daily counts of mentions and archive records with at least one mention

### HORIZON AND INFORMATION CUTOFF

3 Oct - 2 Nov 2026: 31 days
Information available through 2 October

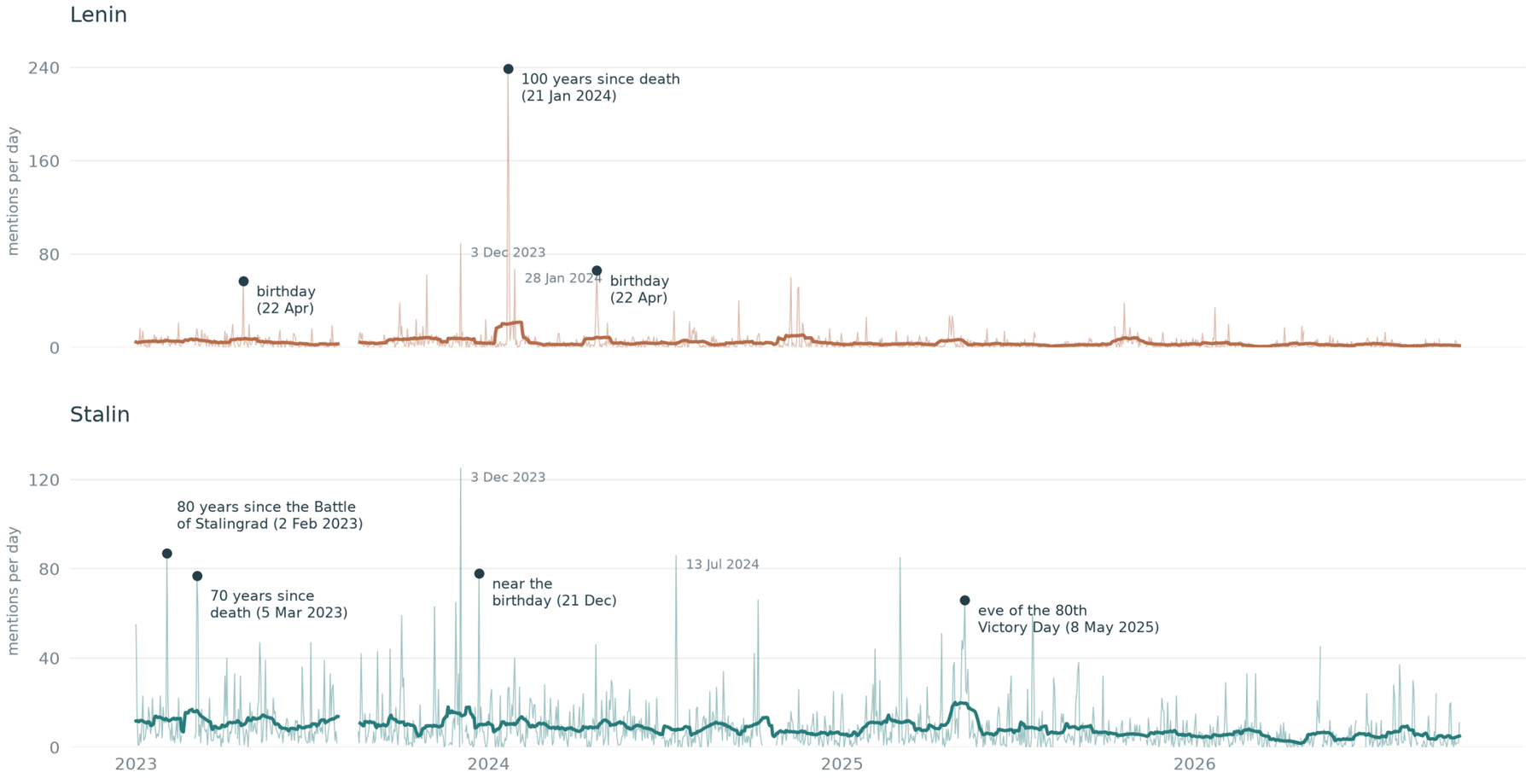
### FORECAST SHAPE

Predictive distributions for integer counts; means and 80% / 95% predictive intervals

Mention counts alone do not measure news quality or establish that a message is propaganda.

# Data: outcomes and forecasting information

GDELT broadcast transcripts, Jan 2023 - Oct 2026; Channel One, NTV, Russia 1 and Russia 24




Labelled: anniversaries from the model calendar; grey: the largest peaks without a calendar date. A coinciding date does not imply a cause.

## Outcomes

Mentions and records with at least one mention
2 figures × 4 channels
× 2 targets = 16 series

## Predictor information

Past archive counts
Channel and time trend
Month and weekday
Known anniversary dates

## 5,334 / 5,484

observed channel-days
Archive gaps are not zeros

Names are screened by context to distinguish person references from named objects and other people.

# Models and the calendar patterns they capture

M1 is the baseline; M3HR extends M3 with partial pooling and a shared calendar-month effect

## Models compared

**M1** — **Constant rates**
Poisson / Binomial baseline

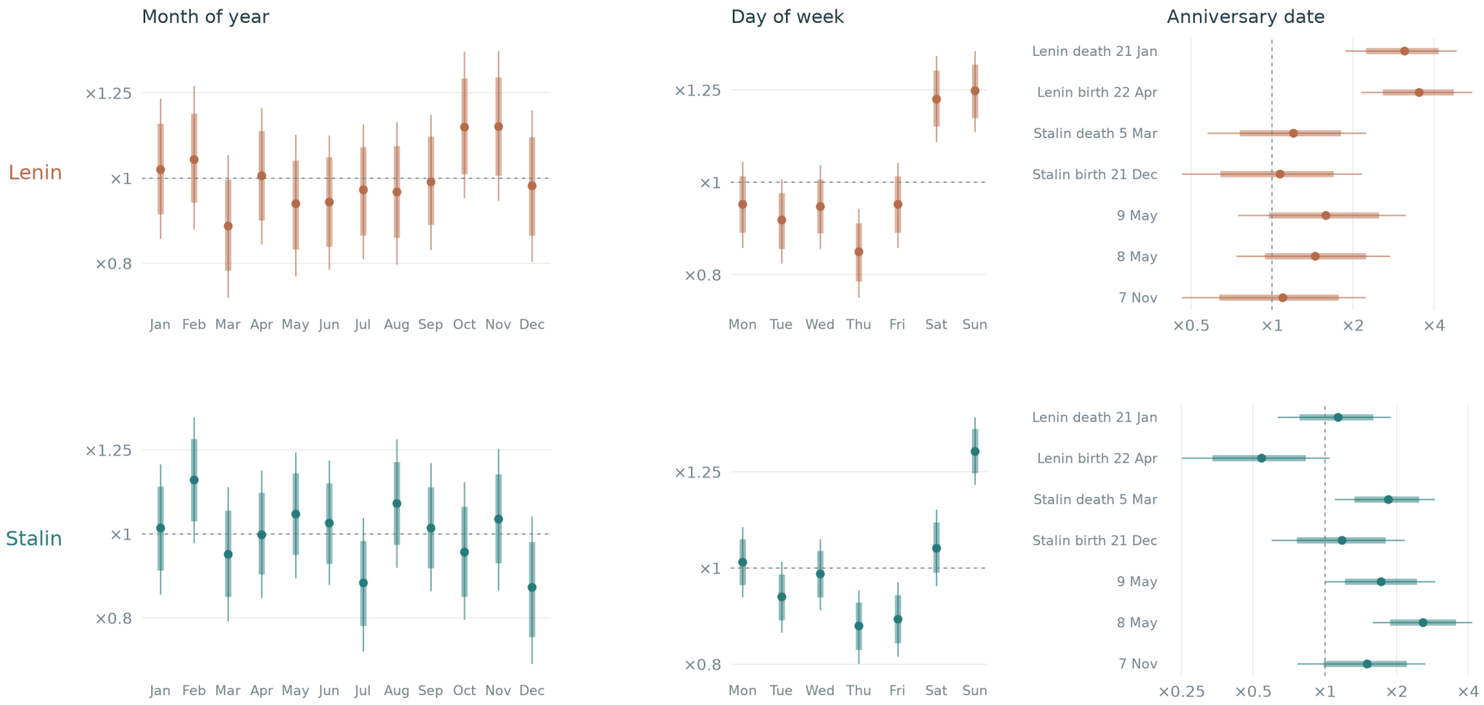
**M2** — **Overdispersion**
Allows bursty daily counts

**M3** — **Calendar + trend**
Month, weekday, anniversaries

**M4** — **Dynamic level**
M3 + random-walk levels

**M3HR** — **Final model**
M3 + estimated shrinkage
+ shared month effect

## Final-model posterior calendar effects



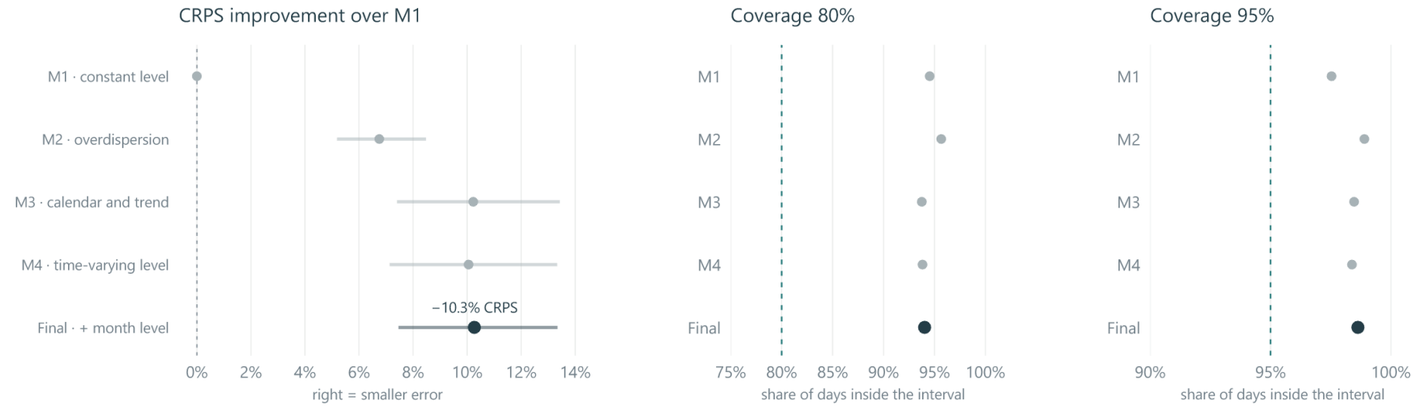

Component "a record mentions the figure" of the final model (beta-binomial, logit). Month effects are shrunk towards a common level. Coefficients are conditional, not causal.

Odds multipliers for a record mentioning each figure.
1 = no effect; bands show 80% and 95% posterior intervals.

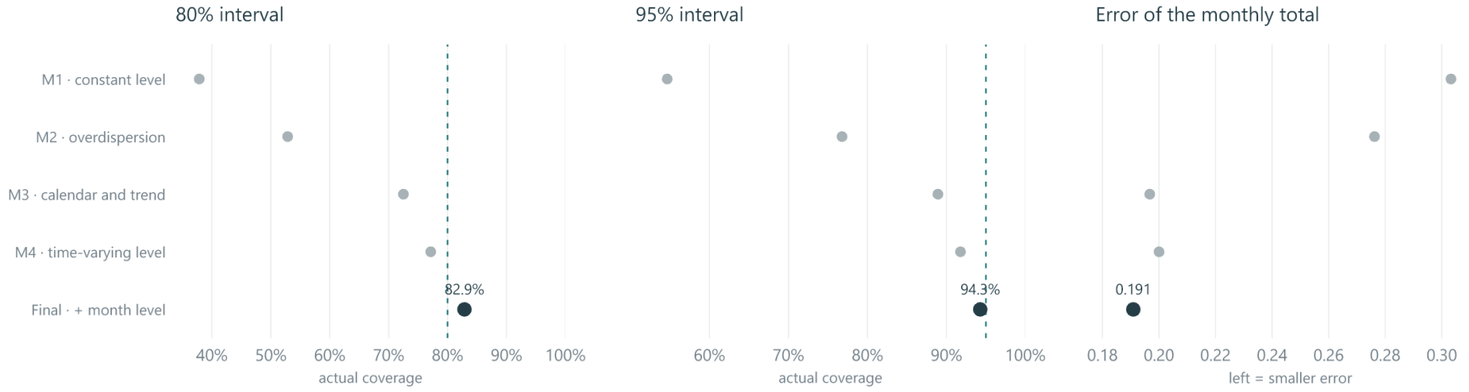
Shared structure: predict archive volume, records with a mention, then total mentions. Monthly intervals use joint trajectories.

# Backtesting results: daily accuracy and monthly uncertainty

21 expanding-window test months, Jan 2025 - Sep 2026; models use the same observations




Bars on the left: 95% block bootstrap (3-month blocks).
Final = M3 + shared month level + estimated shrinkage of month effects; its structure was chosen after the outer test, so the scores are optimistic.




Dashed line - nominal level. The month level shared by channels widens monthly intervals to the nominal level.
Final = M3 + shared month level + estimated shrinkage of month effects; its structure was chosen after the outer test, so the scores are optimistic.

## Daily distributions

**10.3% lower CRPS than baseline M1**

Only 0.046% below M3. Most gain comes from calendar effects.

## Monthly totals

**2.96% lower CRPS than M3**

80% coverage: 82.9%
95% coverage: 94.3%

# Daily forecast: mentions on each channel

M3HR, 3 Oct - 2 Nov 2026; predictive distributions for daily mention counts




A gap in history = an incomplete day. Bands describe future observed counts, including day-to-day variation.

## What is shown?

Grey: observed history
Line: predictive mean
Bands: 80% and 95% predictive intervals

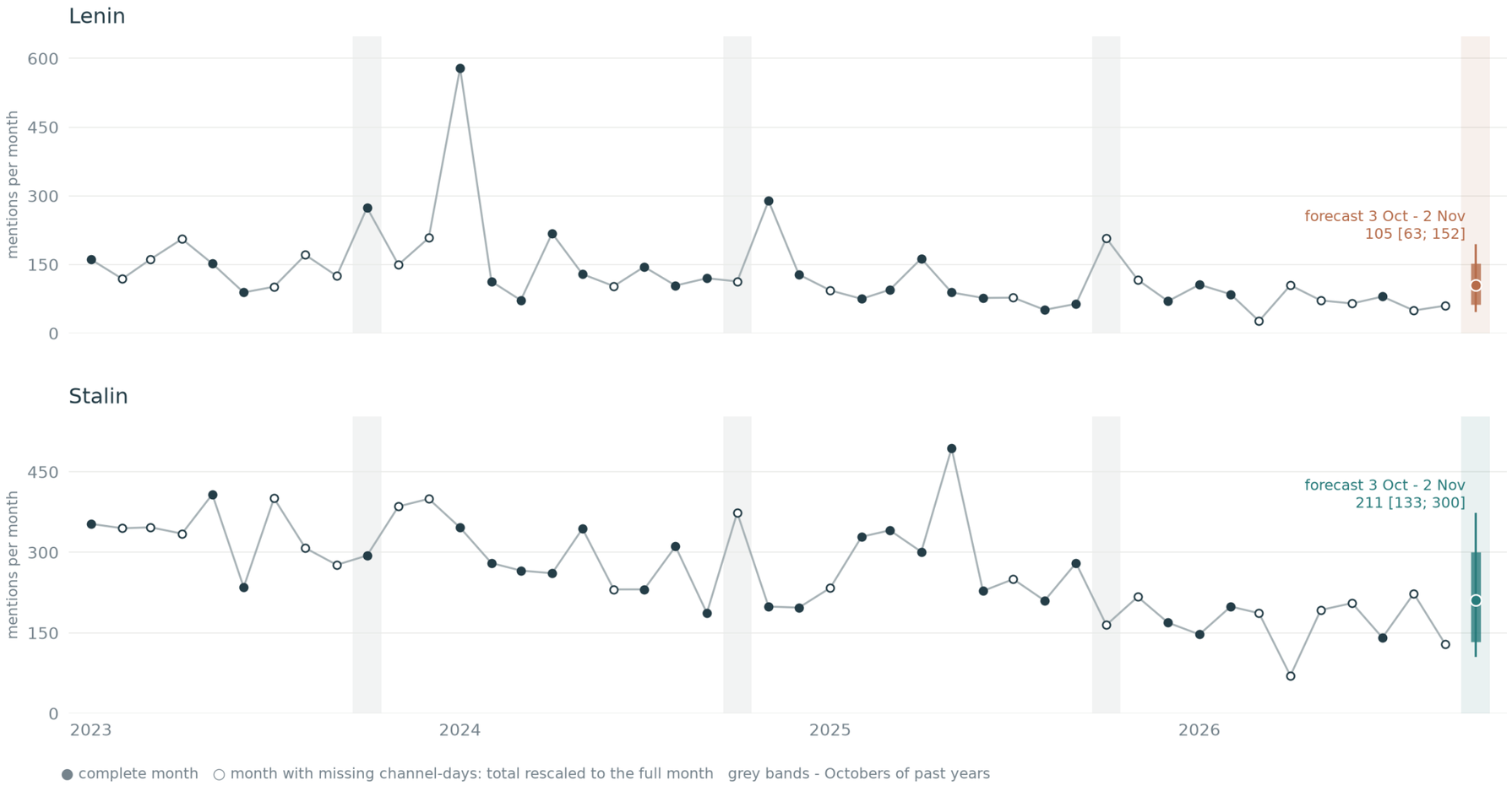
## How to read it

Small daily means coexist with occasional bursts. Intervals include daily count variation and parameter uncertainty.

**Conclusion: daily accuracy is stable; monthly uncertainty improves. Validate M3HR on new forecast origins.**

# M3HR forecast: 3 October - 2 November 2026

Mentions, sum of four channels; point: predictive mean, thick line: 80%, thin line: 95%




## Lenin

**105**

mentions over 31 days

80%: 63-152
95%: 49-192

## Stalin

**211**

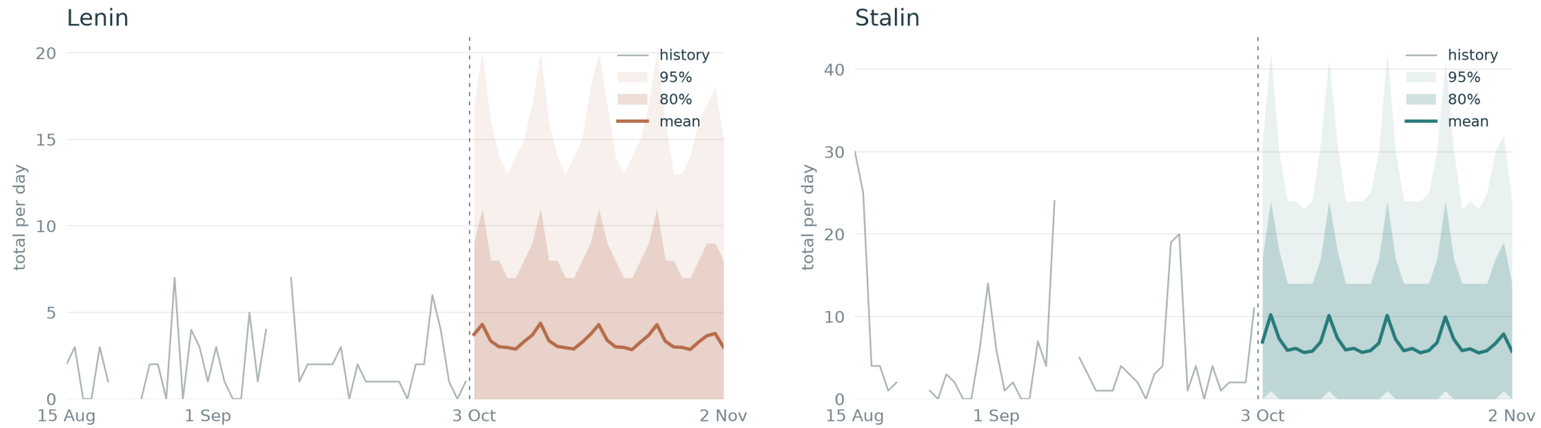
mentions over 31 days

80%: 133-300
95%: 107-371

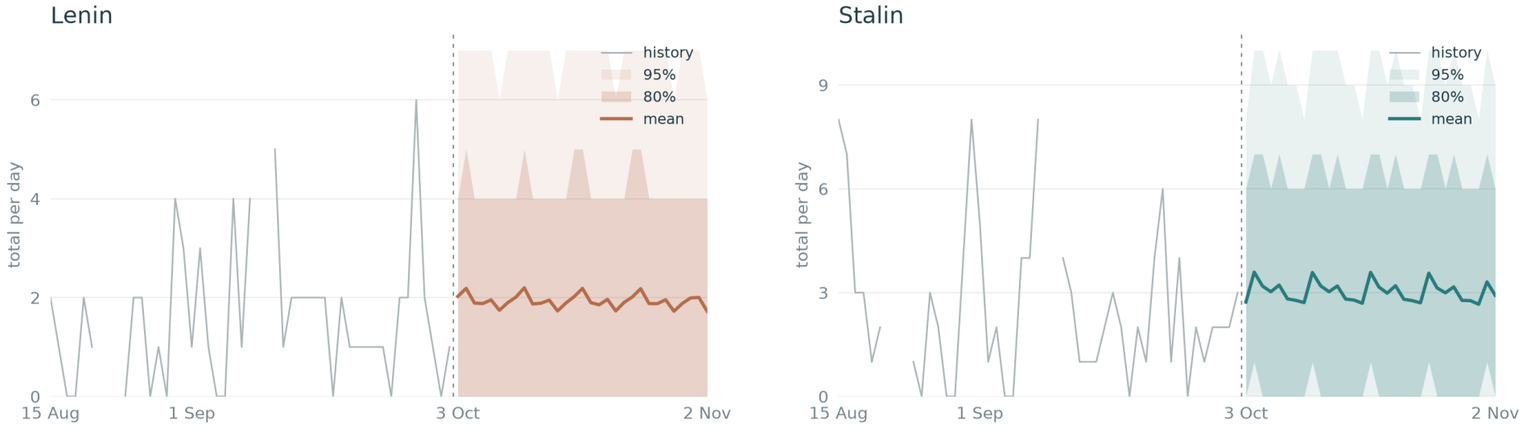
**Daily quality is largely unchanged; monthly uncertainty is better represented. Check future forecast origins.**

# Daily totals across all four channels

M3HR, 3 Oct - 2 Nov 2026; mean and 80% / 95% predictive intervals




Channels are summed within each joint trajectory before computing 80% and 95% intervals. A gap = an incomplete archive day.




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## All mentions

Daily sum across channels.
Left: Lenin
Right: Stalin

## Records with a mention

The second target counts each archive record once, even with repeated mentions.

Channels are summed within each joint trajectory before calculating predictive intervals.